2025 WSMO Speed Round Solutions

SMO Team

SR 1: How many permutations $(a_1, a_2, a_3, a_4, a_5)$ of $(1, 2, 3, 4, 5)$ satisfy

$$a_1 + a_2 = a_4 + a_5?$$

Answer: $\boxed{24}$

Solution: Since $a_1 + a_2 + a_3 + a_4 + a_5 = 15$, we have $a_1 + a_2 = a_4 + a_5 = \frac{15-a_3}{2}$, so a_3 must be odd. If $a_3 = 1$, the remaining numbers split into the pairs $\{2, 5\}$ and $\{3, 4\}$. If $a_3 = 3$, they split into $\{1, 5\}$ and $\{2, 4\}$, and if $a_3 = 5$, they split into $\{1, 4\}$ and $\{2, 3\}$. In each case, there are 2 ways to choose which pair goes first and $2 \cdot 2$ ways to order the numbers within the pairs, giving

$$3 \cdot 2 \cdot 4 = \boxed{24}$$

permutations.

SR 2: Charlie has five consecutive integers and chooses one of them. The sum of her four numbers that she did not choose is 2025. What number did she choose?

Answer: $\boxed{505}$

Solution: Let the five integers be $n-2, n-1, n, n+1, n+2$, which have a sum of $5n$. If Charlie chooses x, then $5n - x = 2025$. Since $n - 2 \leq x \leq n + 2$, we have

$$4n - 2 \leq 2025 \leq 4n + 2 \implies n = 506,$$

meaning $x = 5 \cdot 506 - 2025 = \boxed{505}$.

SR 3: There exists a unique positive integer n such that $n^3 + 6n^2 + 11n + 21$ is a perfect cube. Find the value of n.

Answer: $\boxed{13}$

Solution: Note that

$$n^3 + 6n^2 + 11n + 21 = (n + 2)^3 - (n - 13).$$

If $n < 13$, then the expression is strictly between $(n+2)^3$ and $(n+3)^3$, since $(n+3)^3 - (n+2)^3 = 3n^2 + 15n + 19 > 13 - n$. If $n > 13$, then the expression is strictly between $(n+1)^3$ and $(n+2)^3$, since $(n+2)^3 - (n+1)^3 = 3n^2 + 9n + 7 > n - 13$. So, the expression can only be a perfect cube when $n = 13$, which gives 15^3. So, the answer is $\boxed{13}$.

SR 4: Suppose that three fair six-sided dice are tossed so that the three numbers that they land on are the roots of the monic cubic $P(x) = x^3 + ax^2 + bx + c$. What is the expected value of $200 + 4a + 2b + 4c$?

Answer: $\boxed{60}$

Solution: Let the three numbers be r, s, t. From Vieta's formulas, we have

$$a = -(r + s + t), \quad b = rs + rt + st, \quad \text{and} \quad c = -rst.$$

Since the dice are independent and each has an expected value of $\frac{7}{2}$, we have

$$\mathbb{E}(a) = -\frac{21}{2}, \quad \mathbb{E}(b) = 3\left(\frac{7}{2}\right)^2 = \frac{147}{4}, \quad \text{and} \quad \mathbb{E}(c) = -\left(\frac{7}{2}\right)^3 = -\frac{343}{8}.$$

So, the answer is

$$200 + 4\left(-\frac{21}{2}\right) + 2\left(\frac{147}{4}\right) + 4\left(-\frac{343}{8}\right) = 200 - 42 + \frac{147}{2} - \frac{343}{2} = \boxed{60}.$$

SR 5: Real numbers a and b satisfy

$$\frac{a}{b} + \frac{b}{a} = \frac{a}{b^2} + \frac{b}{a^2} = 10.$$

Then, ab can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{268}$

Solution: Let $s = a + b$ and $p = ab$. From the first equation, we have

$$\frac{a^2 + b^2}{ab} = \frac{s^2 - 2p}{p} = 10 \implies s^2 = 12p.$$

From the second equation, we have

$$\frac{a^3 + b^3}{a^2b^2} = \frac{s^3 - 3ps}{p^2} = 10 \implies s(s^2 - 3p) = 10p^2 \implies 9ps = 10p^2 \implies s = \frac{10p}{9}.$$

Substituting into $s^2 = 12p$, we have

$$\frac{100p^2}{81} = 12p \implies p = \frac{243}{25}.$$

Since $s^2 - 4p = 8p > 0$, a and b are real, giving an answer of $243 + 25 = \boxed{268}$.

SR 6: Suppose $\mathcal{R}_1$ and $\mathcal{R}_2$ are rectangles such that every vertex of $\mathcal{R}_1$ lies on a different side of $\mathcal{R}_2$. The side lengths of $\mathcal{R}_1$ are in a $3 : 4$ ratio, and the side lengths of $\mathcal{R}_2$ are in an $8 : 9$ ratio. Then, the ratio between the perimeters of $\mathcal{R}_1$ and $\mathcal{R}_2$ is $m : n$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{30}$

Solution: WLOG, let the side lengths of $\mathcal{R}_1$ be 3 and 4, and let θ be the angle between a side of $\mathcal{R}_2$ and a side of length 3 of $\mathcal{R}_1$. Each side of $\mathcal{R}_2$ is split by a vertex of $\mathcal{R}_1$ into the legs of two right triangles, so the side lengths of $\mathcal{R}_2$ are

$$3\cos\theta + 4\sin\theta \quad \text{and} \quad 3\sin\theta + 4\cos\theta.$$

By symmetry, we may assume that $9(3\cos\theta + 4\sin\theta) = 8(3\sin\theta + 4\cos\theta)$. This gives $12\sin\theta = 5\cos\theta$, so $\sin\theta = \frac{5}{13}$ and $\cos\theta = \frac{12}{13}$. So, the perimeter of $\mathcal{R}_2$ is

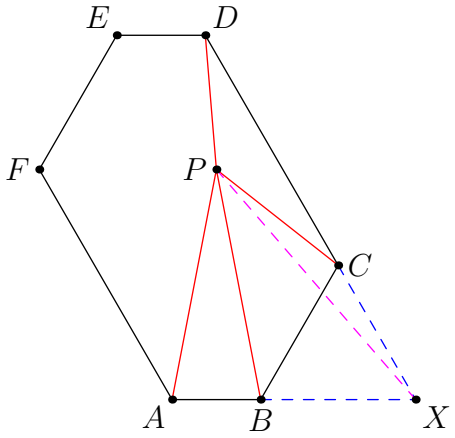
$$2 \cdot 7(\sin\theta + \cos\theta) = 14 \cdot \frac{17}{13},$$

and the perimeter of $\mathcal{R}_1$ is 14, meaning the ratio is $13 : 17 \implies 13 + 17 = \boxed{30}$.

SR 7: Let $ABCDEF$ be an equiangular hexagon with $AB = DE = 4$, $BC = EF = 7$, and $CD = FA$. Point P lies in the interior of the hexagon such that $[APB] = 12\sqrt{3}$, $[BPC] = 14\sqrt{3}$, and $[CPD] = 9\sqrt{3}$. Find the perimeter of the hexagon.

Answer: $\boxed{46}$

Solution:



Let $CD = FA = x$, and let lines AB and CD meet at X. Since the hexagon is equiangular, triangle XBC is equilateral with side length 7. Since $[APB] = 12\sqrt{3}$ and $AB = 4$, the distance from P to line AB is $6\sqrt{3}$, so $[PXB] = \frac{1}{2} \cdot 7 \cdot 6\sqrt{3} = 21\sqrt{3}$. Similarly, the distance from P to line CD is $\frac{18\sqrt{3}}{x}$, so $[PXC] = \frac{63\sqrt{3}}{x}$. Now, since P lies inside the hexagon, we have

$$[PXB] + [PXC] = [XBC] + [BPC] \implies 21\sqrt{3} + \frac{63\sqrt{3}}{x} = \frac{49\sqrt{3}}{4} + 14\sqrt{3} \implies x = 12.$$

So, the perimeter of the hexagon is $2(4 + 7 + 12) = \boxed{46}$.

SR 8: Alice rolls a number t on a fair six-sided die. Bob then rolls t fair twenty-sided dice, getting $a_1, a_2, \ldots, a_t$. Let a be the maximum integer such that 2^a divides $\prod_{i=1}^{t} a_i!$. The expected value of a can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{152}$

Solution: Note that $a = \sum_{i=1}^{t} v_2(a_i!)$. For $k = 1, 2, \ldots, 20$, the values of $v_2(k!)$ are

$$0, 1, 1, 3, 3, 4, 4, 7, 7, 8, 8, 10, 10, 11, 11, 15, 15, 16, 16, 18,$$

which sum to 168. So, each twenty-sided die contributes an expected $\frac{168}{20} = \frac{42}{5}$ to a. Since Alice rolls an expected $\frac{7}{2}$ dice for Bob, we have

$$\mathbb{E}(a) = \frac{7}{2} \cdot \frac{42}{5} = \frac{147}{5} \implies 147 + 5 = \boxed{152}.$$

SR 9: Let $\mathcal{S}$ denote the set of all permutations of the first 6 positive integers. A permutation $(a_1, a_2, \ldots, a_6)$ is chosen uniformly at random from the elements of $\mathcal{S}$. Suppose that there are r distinct values among the remainders when $a_1 + 1, a_2 + 2, \ldots, a_6 + 6$ are each divided by 6. The expected value of r can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Answer: $\boxed{115}$

Solution: Fix a remainder k. None of $a_1 + 1, a_2 + 2, \ldots, a_6 + 6$ leave a remainder of k if and only if $a_i \not\equiv k - i \pmod 6$ for all i. Since the forbidden values $k - 1, k - 2, \ldots, k - 6$ are distinct modulo 6, the number of such permutations is the number of derangements of 6 objects, $D_6 = 265$. So, the probability that k appears as a remainder is $1 - \frac{265}{720} = \frac{91}{144}$. From linearity of expectation, we have

$$\mathbb{E}(r) = 6 \cdot \frac{91}{144} = \frac{91}{24} \implies 91 + 24 = \boxed{115}.$$

SR 10: Complex numbers a, b, c satisfy the following system of equations:

$$\begin{aligned} (a+b)^2 &= ab + 39, \\ (b+c)^2 &= bc + 61, \\ ab + bc + ca &= 38. \end{aligned}$$

Find the sum of all possible values of $(a + b + c)^2$.

Answer: $\boxed{314}$

Solution: Let $s = a + b + c$. Note that $c^2 - ab = c(a + b + c) - (ab + bc + ca) = cs - 38$, so

$$(a+b)^2 - ab = (s-c)^2 - ab = s^2 - 2cs + (cs - 38) = s^2 - cs - 38.$$

So, the first equation becomes $cs = s^2 - 77$. In the same manner, the second equation becomes $as = s^2 - 99$. Since $as + bs + cs = s^2$, we have $bs = 176 - s^2$. Now, let $x = s^2$. Multiplying the third equation by s^2, we have

$$\begin{aligned} 38x &= (as)(bs) + (bs)(cs) + (cs)(as) \\ &= (x - 99)(176 - x) + (176 - x)(x - 77) + (x - 77)(x - 99) \\ &= -x^2 + 352x - 23353 \implies \\ 0 &= x^2 - 314x + 23353 = (x - 121)(x - 193). \end{aligned}$$

Both values give valid solutions, so the answer is $121 + 193 = \boxed{314}$.